Matthew Hase-Liu

Curriculum Vitae

General Information

Email: mhaseliu@mit.edu

Homepage: `https://mhaseliu.github.io/`

Citizenship: US Citizen

Interests: Algebraic geometry and number theory

Appointments

2026–2027 **Research Scientist at MIT**, Cambridge, MA (concurrent with NSF fellowship).
Mentor: Bjorn Poonen

2026–2030 **National Science Foundation Postdoctoral Fellow at MIT**, Cambridge, MA.
Mentor: Bjorn Poonen

Education

2021–2026 **Ph.D. in Math**, *Columbia University*, New York, NY.
Thesis: *Geometry of the moduli space of curves on a hypersurface via the circle method*.
Advisor: Will Sawin.

2017–2021 **A.B. in Math and S.M. in Computer Science**, *Harvard College*, Cambridge, MA.

Awards and Fellowships

2026–2030 NSF Mathematical Sciences Postdoctoral Research Fellowship, MIT

2025 PhD Candidate Recognition Award for Excellence in Teaching, Columbia University

2021–2026 NSF Graduate Research Fellowship, Columbia University

2021 Phi Beta Kappa, Harvard College

2020, 2018 John Harvard Scholar, Harvard College

2019 Certificate of Distinction in Teaching, Harvard University Derek Bok Center for Teaching and Learning

2018 Rank 107.5/4638, William Lowell Putnam Mathematical Competition

2017 Rank 37/280, United States of America Mathematical Olympiad

Professional Service

2026– Journal reviewer for *Algebraic Geometry*, *Essential Number Theory*, *Journal of Number Theory*, and *Proceedings of the 2025 Summer Research Institute in Algebraic Geometry*

2025, 2023 Graduate student mentor for Polymath Jr.

2025– Organizer for Directed Reading Program at Columbia
2023– Graduate student editor for Columbia Journal of Undergraduate Mathematics
2023– Mentor for Directed Reading Program at Columbia
2020 Co-founder and mentor for Mathematics Online Reading Program for High schoolers (MORPH)

Papers, Talks, Conferences, and Seminars

Publications and Preprints

1. Birch's theorem over function fields with quadratically many variables, `https://arxiv.org/abs/2608.27285`.
2. A converse to geometric Manin's conjecture for general low degree hypersurfaces, `https://arxiv.org/abs/2501.12506`, submitted.
3. Terminal singularities of the moduli space of curves on low degree hypersurfaces and the circle method (with Jakob Glas), `https://arxiv.org/abs/2412.14923`, submitted.
4. Non-smoothness of moduli spaces of curves on hypersurfaces (with Amal Mattoo), `https://arxiv.org/abs/2412.04618`, submitted.
5. A geometric approach to functional equations for general multiple Dirichlet series over function fields, `https://arxiv.org/abs/2405.18152`, to appear in **Algebra & Number Theory**.
6. A higher genus circle method and an application to geometric Manin's conjecture, `https://arxiv.org/abs/2402.10498`, to appear in **Algebra & Number Theory**.
7. Sum-product phenomena for planar hypercomplex numbers (with Adam Sheffer), **European Journal of Combinatorics**, Volume 89, Oct 2020, `https://arxiv.org/abs/1812.09547`.

Talks and Posters

1. Johns Hopkins Algebraic Geometry Seminar, 04/26: *Bounding the singular locus of the moduli of curves on a hypersurface*, invited talk.
2. AMS Spring Eastern Sectional, 03/26: *Investigating singularities of moduli spaces with analytic number theory*, invited talk.
3. UIC Algebraic Geometry Seminar, 02/26: *Bounding the singular locus of the moduli of curves on a hypersurface*, invited talk.
4. Stanford Number Theory Seminar, 01/26: *Unusual applications of analytic number theory to algebraic geometry*, invited talk.
5. University of Michigan Algebraic Geometry Seminar, 09/25: *Applications of analytic number theory to the geometry of moduli spaces*, invited talk.
6. Summer Research Institute in Algebraic Geometry, 07/25: *Moduli spaces of curves on low degree hypersurfaces and the circle method*, poster.
7. Institut Mittag-Leffler, 07/25: *Terminality of moduli spaces of curves on low degree hypersurfaces via the circle method I*, invited talk.
8. AMS New England Graduate Student Conference, 04/25: *The geometry of moduli spaces of curves on varieties via analytic number theory*, invited talk.
9. Tufts Algebra, Geometry, and Number Theory Seminar, 04/25: *Terminality of the moduli space of curves on low degree hypersurfaces and the circle method*,

invited talk.

10. University of Maryland Number Theory and Representation Theory Seminar, 03/25: *Investigating singularities of moduli spaces with analytic number theory*, invited talk.
11. Philadelphia Area Number Theory Seminar, 10/24: *Geometric aspects of general multiple Dirichlet series over function fields*, invited talk.
12. Harvard–MIT Algebraic Geometry Seminar, 10/24: *A circle method for algebraic geometers*, invited talk.
13. Penn State Algebra and Number Theory Seminar, 04/24: *A higher genus circle method*, invited talk.
14. Enumerative Geometry and Arithmetic, 3/24: *A higher genus circle method and an application to geometric Manin's conjecture*, poster.
15. Monodromy and Its Applications, 12/23: *Functional equations for multiple Dirichlet series over function fields*, contributed talk.
16. Midwest Arithmetic Geometry and Number Theory Conference Series, 10/23: *The mapping space of a smooth projective curve to a smooth hypersurface of low degree*, poster.
17. Combinatorial and Additive Number Theory, 5/19: *Sum-product phenomena for planar hypercomplex numbers*, invited talk.

Conferences and Workshops

1. AMS Spring Eastern Sectional, 03/26: *Topology and Arithmetic of Moduli Spaces of Curves*.
2. Summer Research Institute in Algebraic Geometry, 07/25.
3. Institut Mittag-Leffler, 07/25: *Full circle: 100 years of the circle method*.
4. AIM workshop, 04/25: *Moments in families of L-functions over function fields*.
5. AMS New England Graduate Student Conference, 04/25.
6. AIM workshop, 11/24: *Nilpotent counting problems in arithmetic statistics*.
7. Enumerative Geometry and Arithmetic, 3/24.
8. Monodromy and Its Applications, 12/23.
9. Midwest Arithmetic Geometry and Number Theory Conference Series, 10/23.
10. SLMath summer school, 7/23: *Introduction to derived algebraic geometry*.
11. Second JNT biennial conference, 7/22.
12. Anabelian days down in Georgia, 4/22.
13. PCMI summer school, 7/21: *Quadratic forms, Milnor K-theory, and arithmetic*.
14. Combinatorial and Additive Number Theory, 5/19.

Organized Seminars

1. *Chabauty–Coleman's method*, learning seminar, Fall 2025.
2. *Exponential sums and equidistribution*, learning seminar, Spring 2025.
3. *Function field arithmetic and geometry*, learning seminar, Spring 2024.
4. *Cohomology and analytic number theory over function fields*, learning seminar, Fall 2022.
5. *Abelian reasons and a variety of examples to care about abelian varieties*, learning seminar, Spring 2022.
6. *DWIC: DG With Infty Categories Seminar*, learning seminar, Fall 2021.

Teaching Experience

2024 **Calculus I instructor**, *Columbia University*.

2021– **Teaching assistant**, *Columbia University*.

2019–2020 **Course assistant**, *Harvard College*.

Computer Skills

Python, Java, Linux, SageMath, PyTorch, Lean (learning)

Languages

English (native), Japanese (fluent)